# Ovarian Cancer Survival Prediction Using Explainable AI

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# Contents

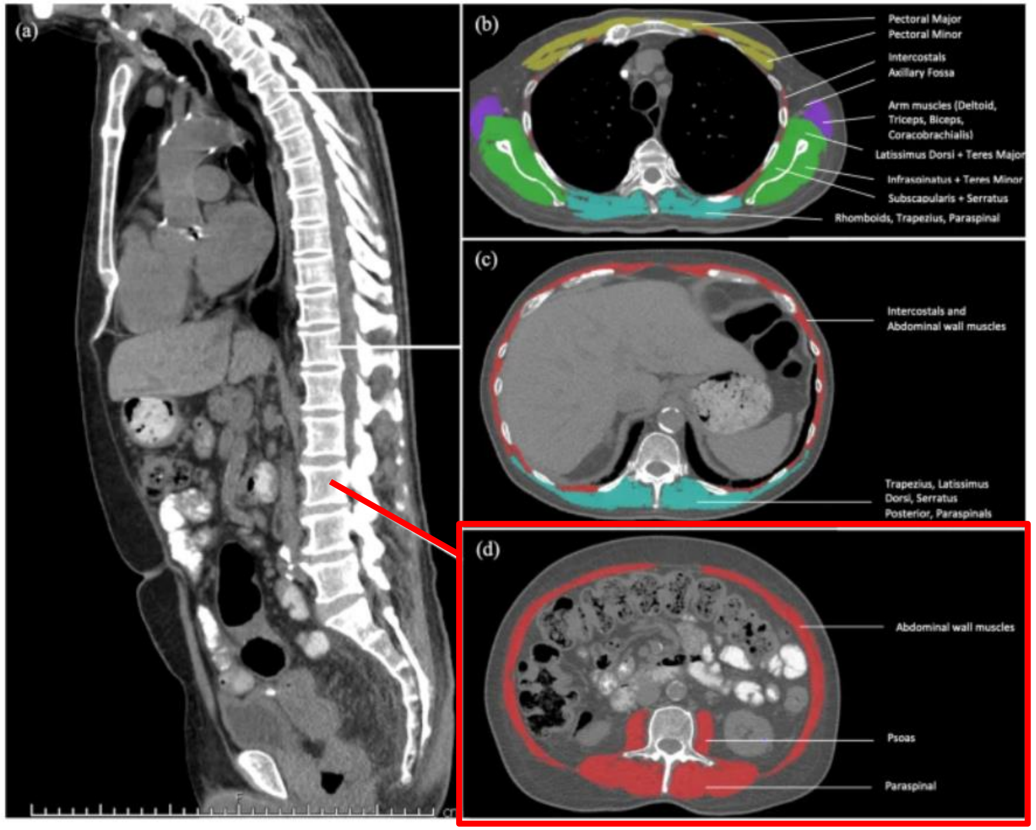
# 1.
# Background and Motivation

# Definition of Skeletal Muscle Index (SMI)

## 1. Skeletal Muscle Index (SMI)

- SMI is a **standardized metric of skeletal muscle mass.**
- Derived from **L3 CT images** using a Hounsfield Unit (HU) range of **-29 to +150**, and serves as a validated proxy for total body skeletal muscle mass (Mourtzakis et al., 2008; Shen et al., 2004).

$$\mathrm{SMI} = \frac{\text{Muscle Area (cm}^2\text{)}}{\text{Height}^2\ (\text{m}^2)}$$




**Single CT slice** at L3 vertebra

Source: Arayne et al. *BMC Cancer* (2023); 23:56.

# Prognostic Meta-analytic Evidence on Pre-treatment SMI

1. **Low SMI**

- Meta-analysis of 7,843 patients with solid tumors from 38 studies (Shachar et al., 2016).
- Associated with **low Overall survival (HR = 1.44) and Cancer-specific survival (HR = 1.93).**

**2. Further Evidence**

Meta-analyses confirmed in specific cancers:

- **Ovarian** (Tranoulis et al., 2022).
- **Hepatocellular** (Xu et al., 2020).
- **Gastric** (Guo et al., 2025).

# Prevalence and Prognostic Significance of Low SMI and SMD in Cancer Patients

**McGovern et al. (2021)**

- Systematic review of **160 studies** including **42,000+ cancer patients.**
- Found that **low SMI and SMD are highly prevalent** across tumor types and stages.
- Prevalence of low SMI/SMD ranged from **35% to 50%**, regardless of threshold, cancer type, or disease stage.
- Suggests that **poor muscle quantity (SMI)** and **quality (SMD)** are **endemic at diagnosis** in cancer patients.

# Traditional Survival Model

- The Cox Proportional Hazards (CPH) model estimates **the risk of an event (e.g., death) over time based on patient covariates.**
- It assumes that the hazard function is a product of a baseline hazard and a log-linear function of covariates.
- As a **semi-parametric model**, it does not require explicit specification of the baseline hazard function.
- However, it has the following limitations:

| Limitation | Description |
|---|---|
| **Linearity assumption** | Assumes a linear relationship between covariates and log hazard (may not hold in practice) |
| **Proportional hazards assumption** | Assumes that hazard ratios between groups are constant over time, which is often violated in dynamic clinical risks |
| **Limited in handling high-dimensional and interaction effects** | Performance deteriorates in the presence of many variables, nonlinear patterns, or complex feature interactions |

# Advancing Survival Prediction: From Cox to AI-based Survival Model

| Aspect | Traditional Cox Models | AI-Based Survival Models |
|---|---|---|
| **Assumptions** | Linear effects, proportional hazards | No assumption of linearity or proportional hazards |
| **Handling of time** | Static risk estimation | Time-dependent, patient-specific survival prediction |
| **High-dimensional data** | Limited capacity, requires manual variable selection | Handles high-dimensional data (e.g., radiomics, genomics) automatically |
| **Interactions between variables** | Typically ignored or simplified | Can model complex, nonlinear interactions |
| **Model interpretability** | Direct interpretation of coefficients | Requires XAI techniques (e.g., SHAP) for interpretability |

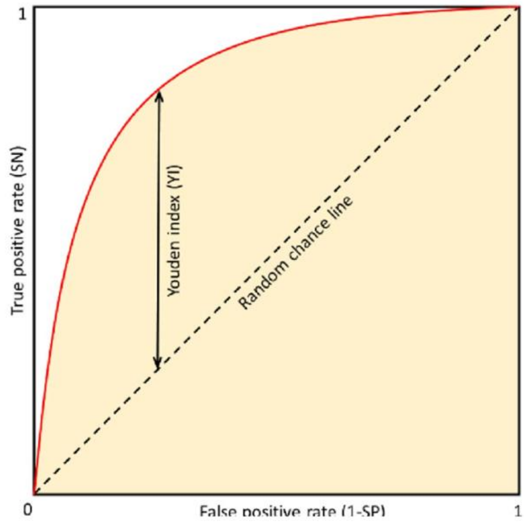
# The Importance of Standardized Skeletal Muscle Cutoffs in Oncology Prognosis

A 2025 subanalysis of the SORAMIC clinical trial evaluated how **14 SMI** and **7 SMD** cutoff definitions influenced sarcopenia/myosteatosis prevalence and survival prediction in 179 liver cancer patients.

- Sarcopenia prevalence ranged from **8.9% to 69.8%**
- Only **3 SMI and 1 SMD** cutoffs showed significant OS prediction (HR ~1.5)
- Most definitions failed to predict prognosis

**These findings highlight the urgent need to establish cancer-specific, standardized cutoff values to improve clinical applicability of body composition metrics.**

# Traditional Statistical Methods for Determining Cut-off Values






## Common Methods

- **ROC Curve Analysis**

  Identifies the point maximizing sensitivity and specificity

  → using **Youden's Index**

- **Optimal Stratification**

  Determines the cut-off that best separates survival curves (e.g., via log-rank test)

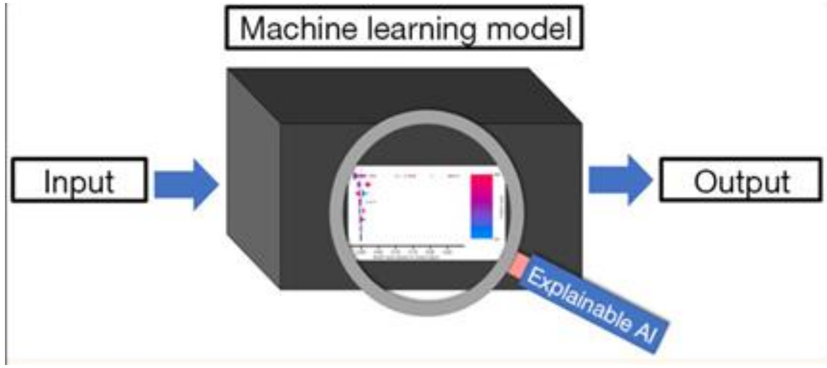
- **Tertile or Quartile Split**

  Uses distribution-based grouping (e.g., lowest 25% vs. others)

# Explainable Artificial Intelligence (XAI)

## Definition

- **XAI aims to provide insight into the inner workings of "black-box" models by generating quantitative and visual explanations of how predictions are made.**




Source: Ladbury et al., Transl Cancer Res, 2022.

## Key Functions of XAI

- **Quantifies** how much each feature contributes to a specific prediction.
- **Visualizes** nonlinear relationships and feature interactions(SHAP dependence plots).
- **Explains** predictions at both **global** (entire population) and **individual** (per patient) levels.

# Explainable machine learning can outperform Cox regression predictions and provide insights in breast cancer survival

## Dataset

- Data: Netherlands Cancer Registry (36,658 patients with non-metastatic breast cancer).
- Variables: demographics, tumor size, nodal status, and receptor subtype, and surgery type.

## Preprocessing

- Basic preprocessing was applied, including imputation for missing values, derivation of clinical features (e.g., lymph node ratio, hormone receptor subtypes), and selection of key variables based on clinical relevance and algorithmic ranking.

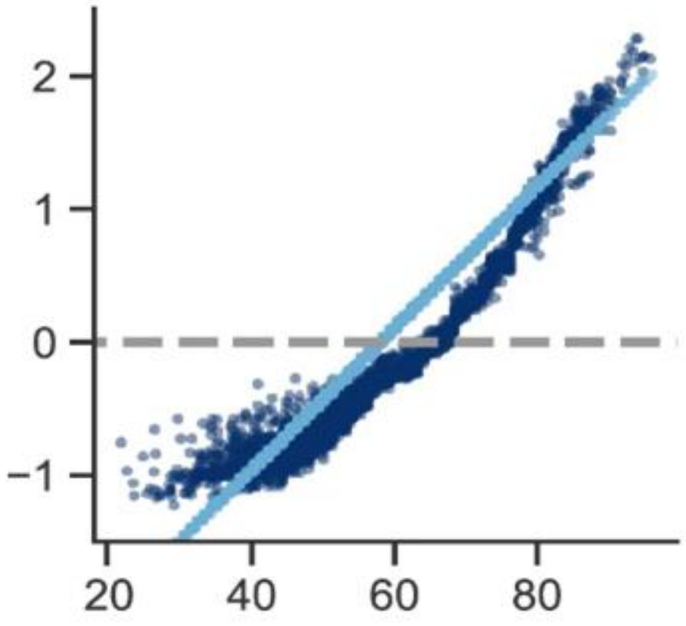
| Model | C-index (Test Set) |
|---|---|
| Cox Proportional Hazards | 0.63 - 0.64 |
| Random Survival Forest | |
| Survival SVM | |
| Extreme Gradient Boosting | 0.73 （$p < 0.0001$） |

## Explainability with SHAP

- **Feature importance ranking**
- **Dependence plots**
  → Revealed **nonlinear patterns** and **turning points** in risk
- **Interaction effects**

## Threshold Identification Strategy

- **SHAP-based turning points** identified risk thresholds, e.g.:
  - Age ~65: SHAP value shifts from negative to positive
  - Tumor size plateaued risk beyond ~45 mm
  - Positive lymph nodes showed risk increase only after ≥1




**(a)** *age*

SHAP feature dependence plots can help us **identify important turning points** for the different features.

# Motivation

- **SMI** is established prognostic clinical indicator in cancer.
- However, **current cut-off values vary** and lack reproducibility.
- Traditional methods use **linear thresholds**, limiting clinical utility.
- **While XAI enables interpretation of model outputs, its use in deriving data-driven thresholds for SMI remains unexplored.**

➤ There is a need for a **data-driven, explainable framework** to identify and validate clinically meaningful thresholds.

# Aim

- **Construct and evaluate survival models** using pre-treatment SMI and clinical covariates.
- **Analyze feature importance and non-linear patterns** using explainable AI (e.g., SHAP).
- **Assess the prognostic impact** of clinical risk factors and discuss methodological challenges in threshold derivation.

# 2.
# Study Design and Methodology

# Dataset Overview

### Source

The dataset was collected from the Department of Radiation Oncology at MacKay Memorial Hospital, Tamsui Branch.

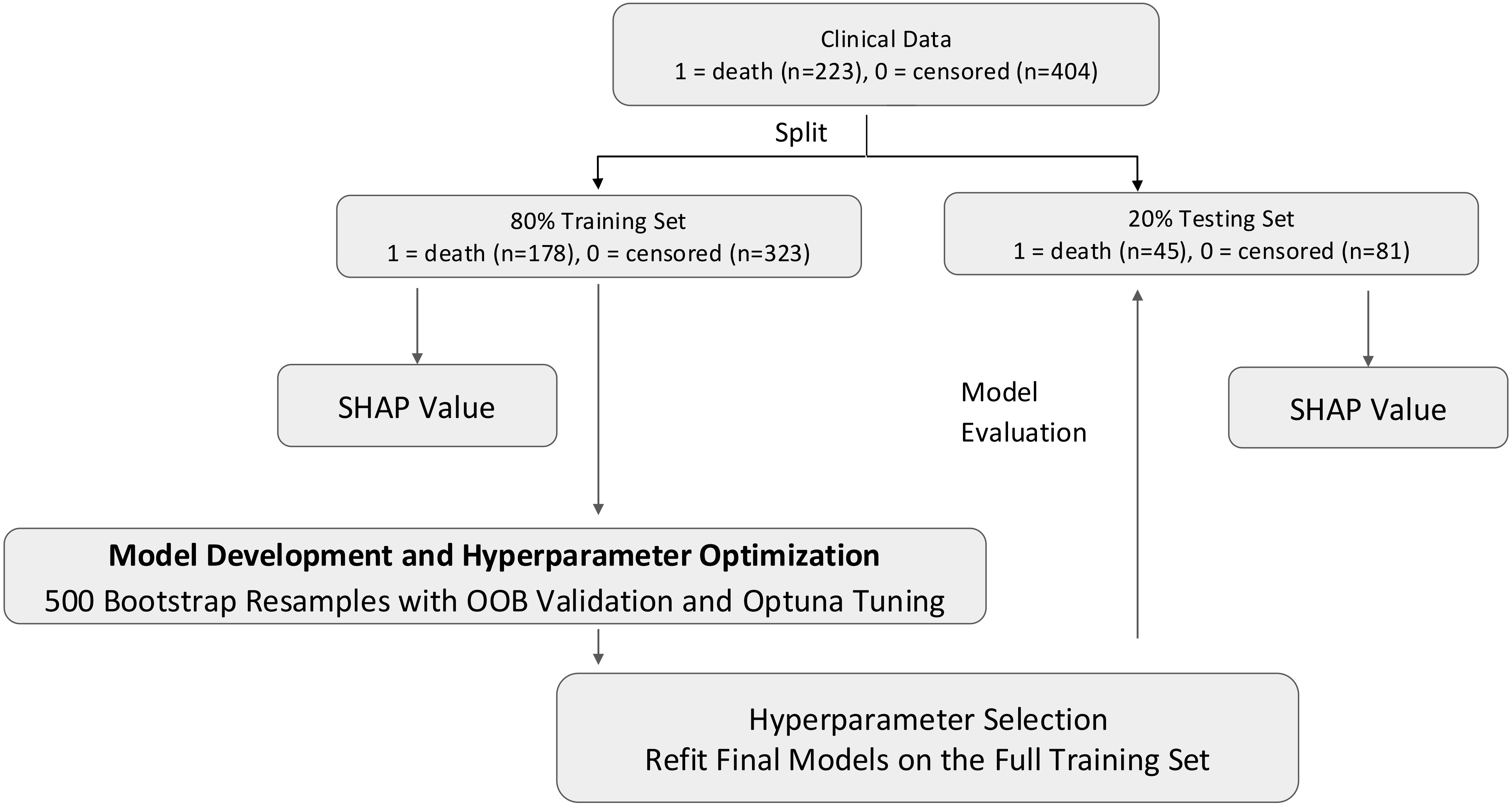
### Study Population

- **627 patients** diagnosed with **ovarian cancer**
- All patients underwent **pre-treatment assessment prior to surgery and chemotherapy**

### Prediction Targets: Overall Survival (OS)

- **Event**: 1 = death (n=223), 0 = censored (n=404)
- **Time**: Duration from diagnosis to death or last follow-up (in years)

# Workflow


Clinical Data
1 = death (n=223), 0 = censored (n=404)
Split
80% Training Set
1 = death (n=178), 0 = censored (n=323)
20% Testing Set
1 = death (n=45), 0 = censored (n=81)
SHAP Value
Model
Evaluation
SHAP Value
Model Development and Hyperparameter Optimization
500 Bootstrap Resamples with OOB Validation and Optuna Tuning
Hyperparameter Selection
Refit Final Models on the Full Training Set

# Pre-treatment Features

- **Demographics & Body Composition:**

Age, Body Mass Index (BMI), Skeletal Muscle Index (SMI)

- **Inflammatory Markers:**

Serum albumin (Alb), Neutrophil-to-Lymphocyte Ratio (NLR), Platelet-to-Lymphocyte Ratio (PLR)

- **Clinical Risk Indicators:**

Malignant ascites (1 = malignant, 0 = non-malignant or none),

FIGO stage grouping (0 = Stage I/II, 1 = Stage III/IV),

ECOG (0 = fully active, 1 = ambulatory; all patients were ECOG 0–1),

Grade (1 = well diff., 2 = mod diff., 3 = poor diff.),

Resec_status (0 = no gross residual, 1 = ≤1 cm, 2 = >1 cm),

Histological cell type (Cell_2: 0 = serous/undifferentiated, 1 = others)

# Model

| **Model** | Mechanism | Notes |
|---|---|---|
| **Cox Proportional Hazards** | Linear regression model that estimates the log-hazard function with proportional hazards assumption. | • Interpretable.<br>• Assumes proportional hazards. |
| **Random Survival Forest (RSF)** | Ensemble of decision trees built on bootstrapped samples; splits nodes by **maximizing survival difference using log-rank** test. | • Handles nonlinearity and censoring.<br>• Outputs survival curves |
| **XGBoost Survival Embedding** | Modified XGBoost with **Cox loss function**; maps features to sparse high-dimensional embeddings for modeling nonlinearity. | • Reduces noise and dimensionality.<br>• Models nonlinear effects |

| **Model** | Mechanism | Notes |
| --- | --- | --- |
| **Survival Support Vector Machine (Survival SVM)** | Extends SVM to censored data by **optimizing a ranking loss** that respects survival ordering | • Focuses on risk ranking.<br>• Good for small datasets |
| **DeepSurv** | Deep feedforward neural network that outputs a nonlinear **log-risk function** via a single linear node. | • Learns nonlinear risk.<br>• Outputs relative risk scores |

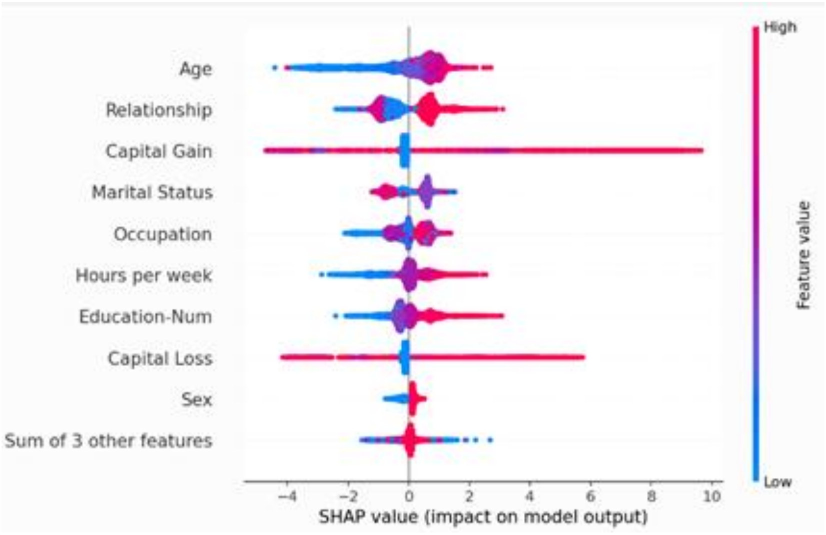
# Evaluation metrics

| Metric | Aspect of Performance | Model Output Type | Description |
|---|---|---|---|
| **C-index** | Discrimination | Time-independent | Agreement between the predicted vs. the observed order of the events. |
| **Time-dependent AUC** | Discrimination | Time-dependen | Reflects the probability that a patient who experiences the event **before a given time** is assigned a **higher risk score** than one who survives longer or is censored. |
| **Calibration plot** | Calibration | Time-dependent | Plot of the event probability values predicted by a model against the observed event probabilities. |

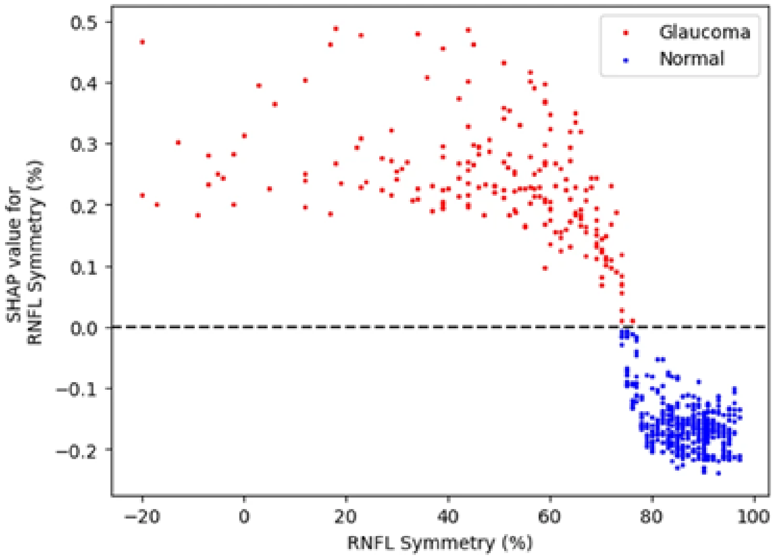
# SHAP (SHapley Additive exPlanations)

- It quantifies each feature's contribution to predictions, enhancing transparency in complex models.

Beeswarm Plot

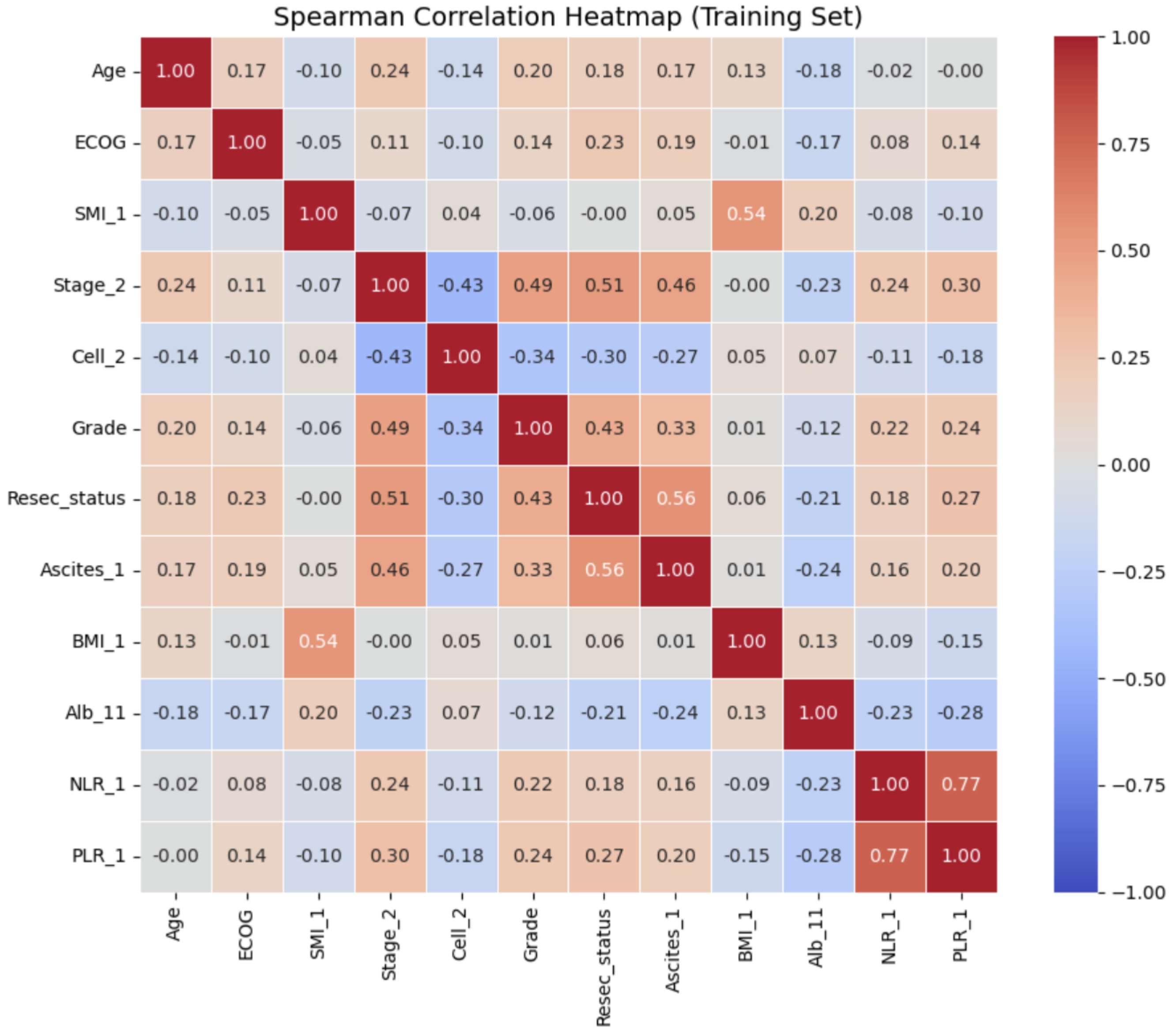



Dependence Plot



# 3.
# Result

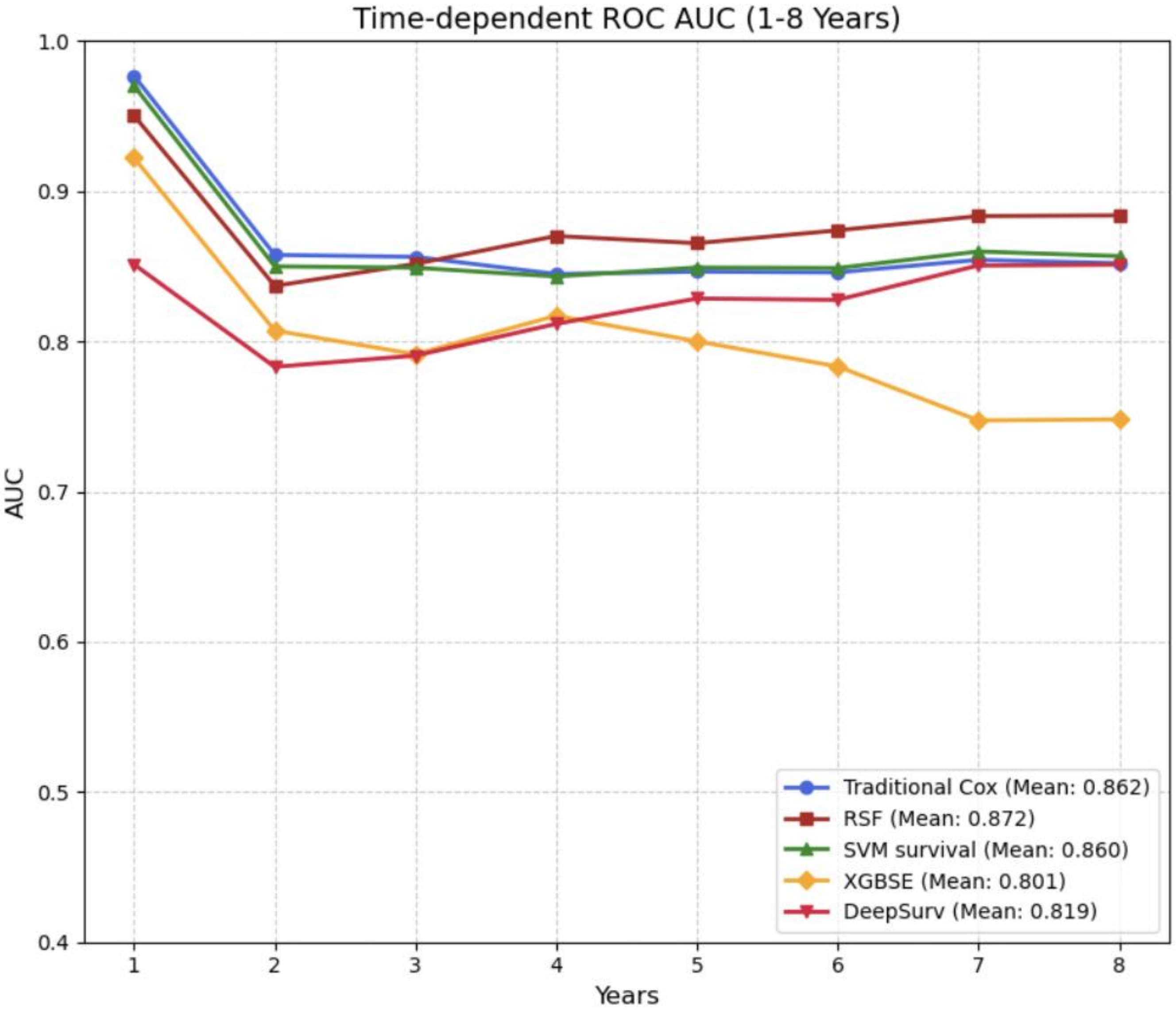
# 3.1 Spearman Analysis



# 3.2 Survival Model Performance

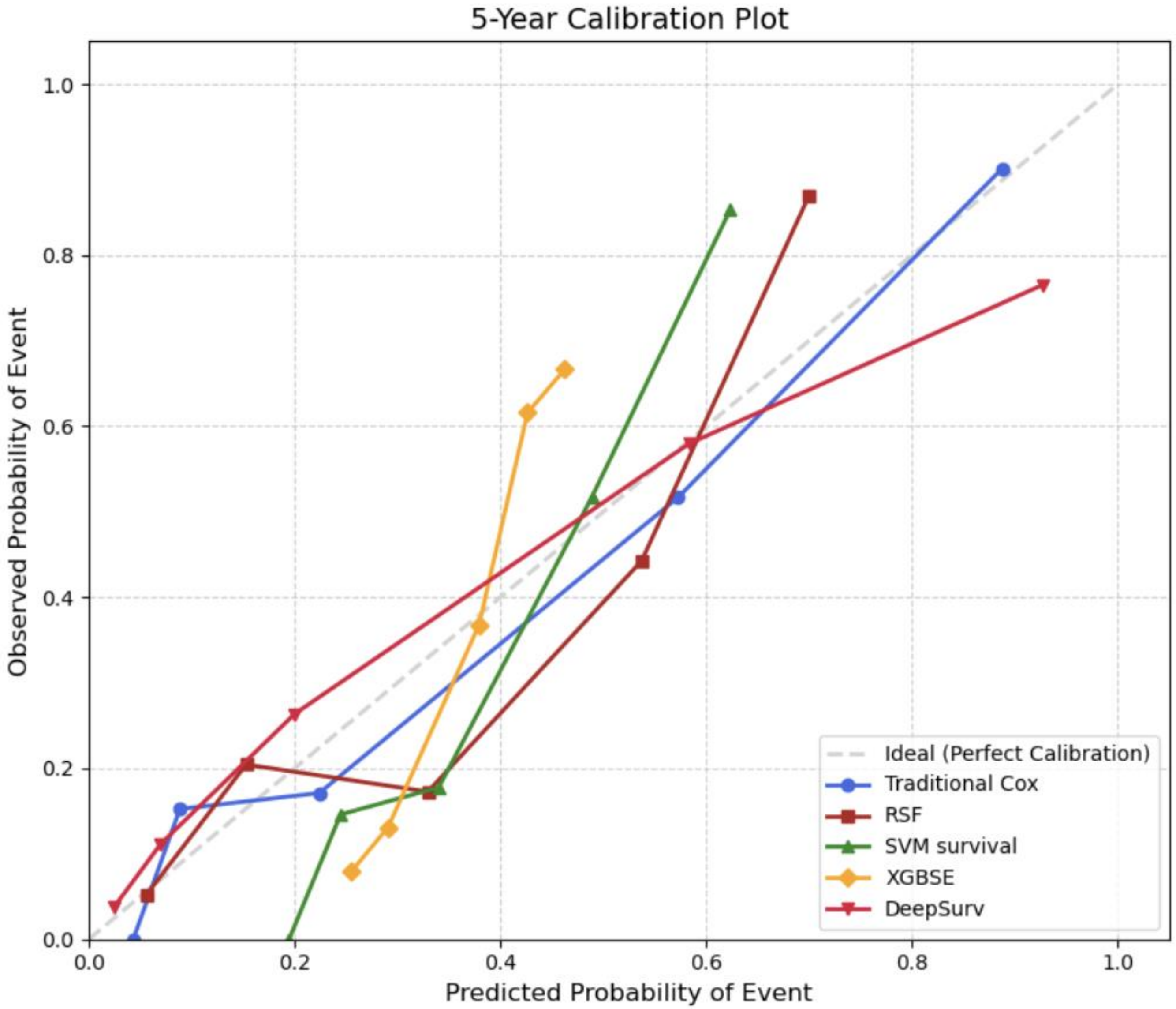
| Model | Boostrap (mean ± std) | Train C-index (Best Hyperparameter Retrain) | Test C-index |
| --- | --- | --- | --- |
| **Cox** | | 0.8414 | 0.8256 |
| **RSF** | 0.8328 ± 0.0184 | **0.8590** | **0.8277** |
| **SVM Survival** | 0.8319 ± 0.0188 | 0.8430 | 0.8246 |
| **XGBSE** | 0.7903 ± 0.0243 | 0.8080 | 0.7669 |
| **DeepSurv** | 0.8107 ± 0.0257 | 0.8284 | 0.7847 |

# Time-dependent ROC




| Model | Mean ROC |
|---|---|
| **Cox** | 0.8618 |
| **RSF** | **0.8715** |
| **SVM Survival** | 0.8601 |
| **XGBSE** | 0.8008 |
| **DeepSurv** | 0.8186 |

# Calibration Plot



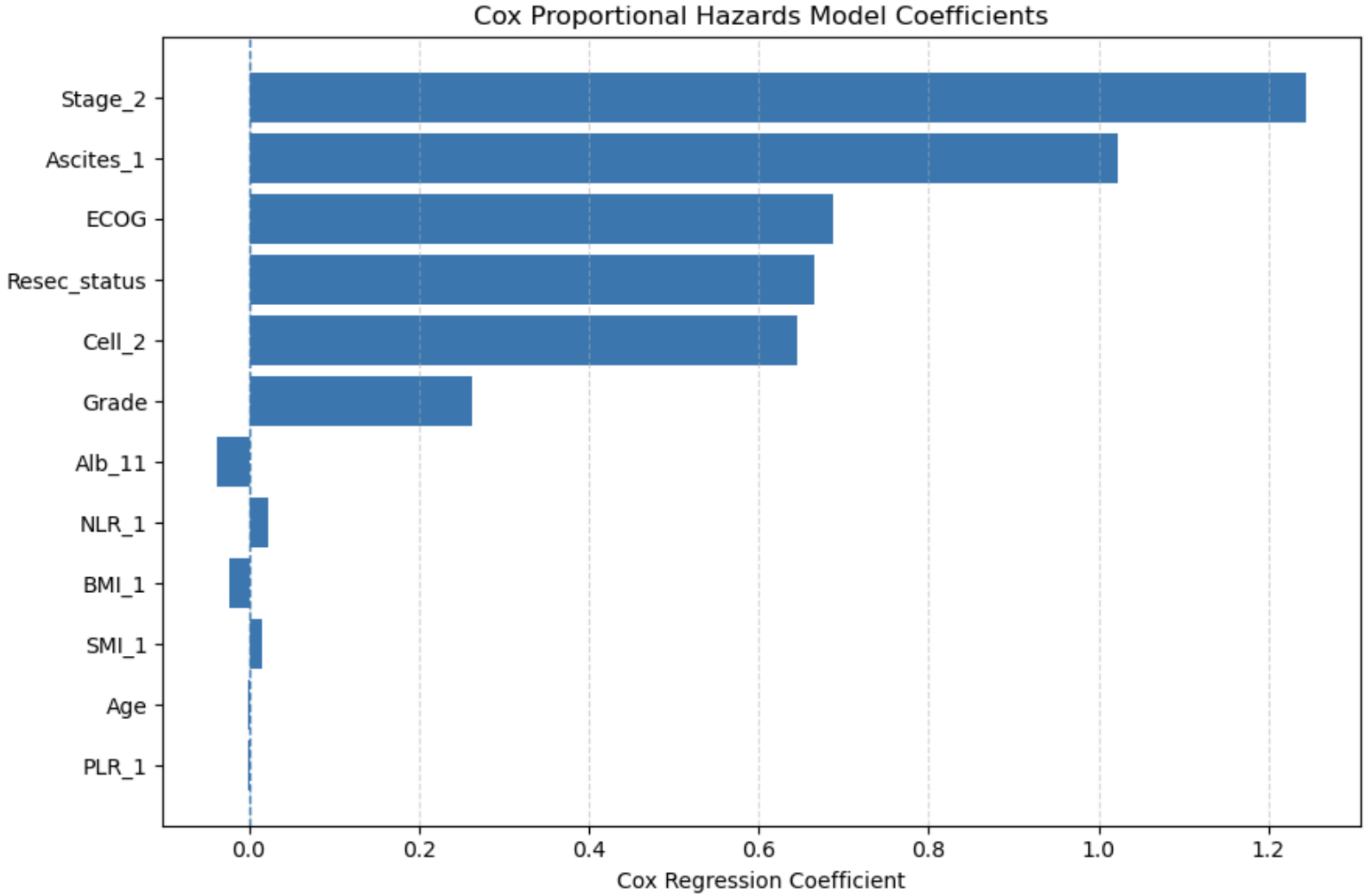

| Model | 5 years Brier Score |
|---|---|
| **Cox** | 0.1575 |
| **RSF** | **0.1508** |
| **SVM Survival** | 0.1677 |
| **XGBSE** | 0.1955 |
| **DeepSurv** | 0.1670 |

# 3.3 Comparative Performance of Survival Models: Test Set 2000 Bootstraps

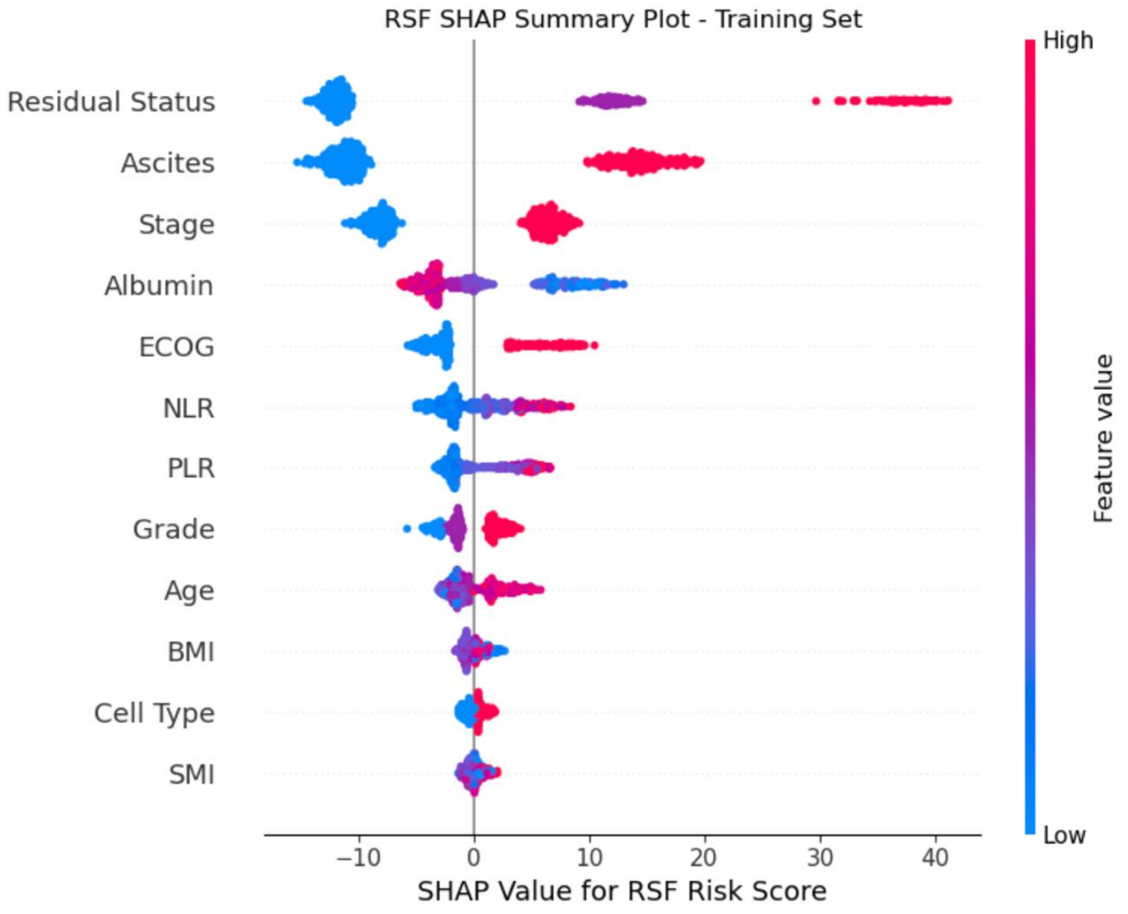
| Model | C-index<br><br>Boostrap Mean | C-index<br><br>95% CI lower | C-index<br><br>95% CI upper |
|---|---|---|---|
| **Cox** | 0.8280 | 0.7615 | 0.8825 |
| **RSF** | 0.8277 | 0.7701 | 0.8837 |
| **SVM Survival** | 0.8244 | 0.7625 | 0.8794 |
| **XGBSE** | 0.7670 | 0.6994 | 0.8326 |
| **DeepSurv** | 0.7846 | 0.7171 | 0.8492 |

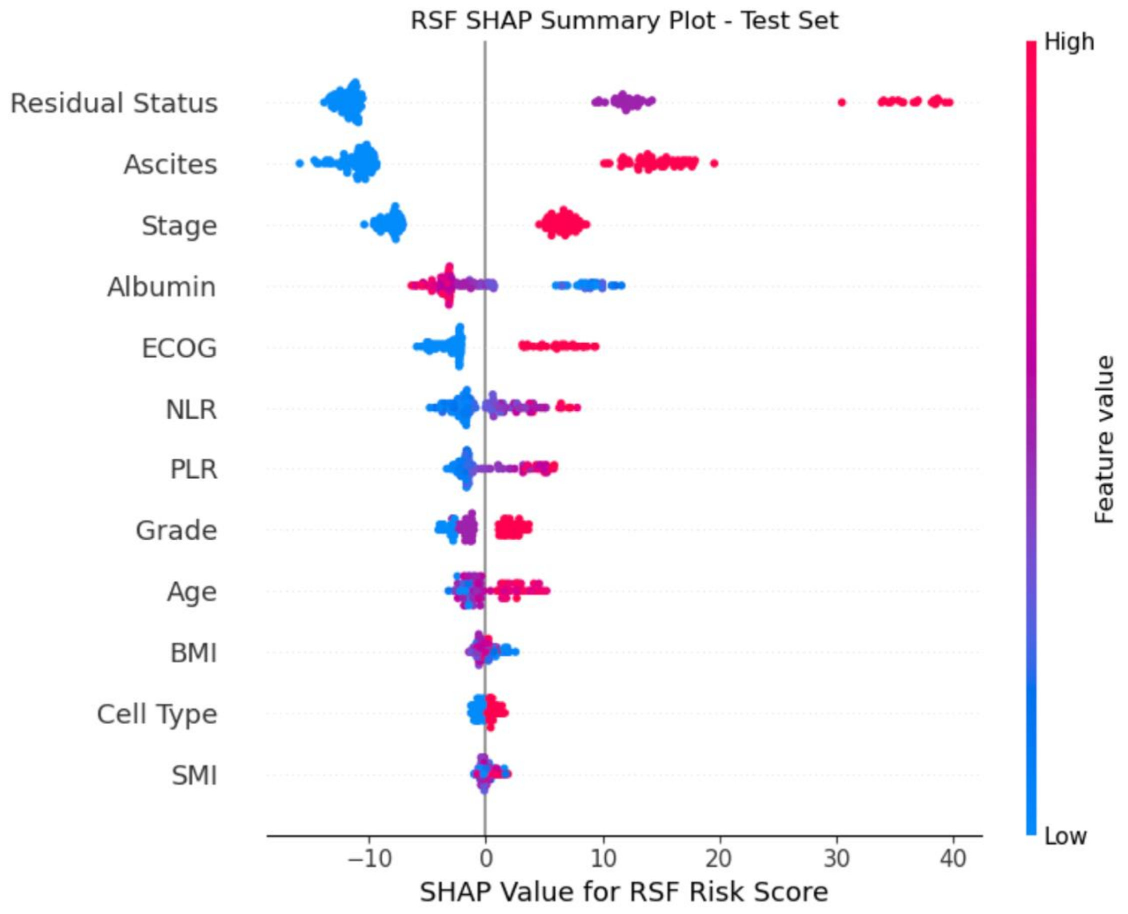
**Pairwise C-index comparisons were performed using paired bootstrap resampling with Holm adjustment; no statistically significant differences remained after correction.**

# 3.4 Explainability Analysis

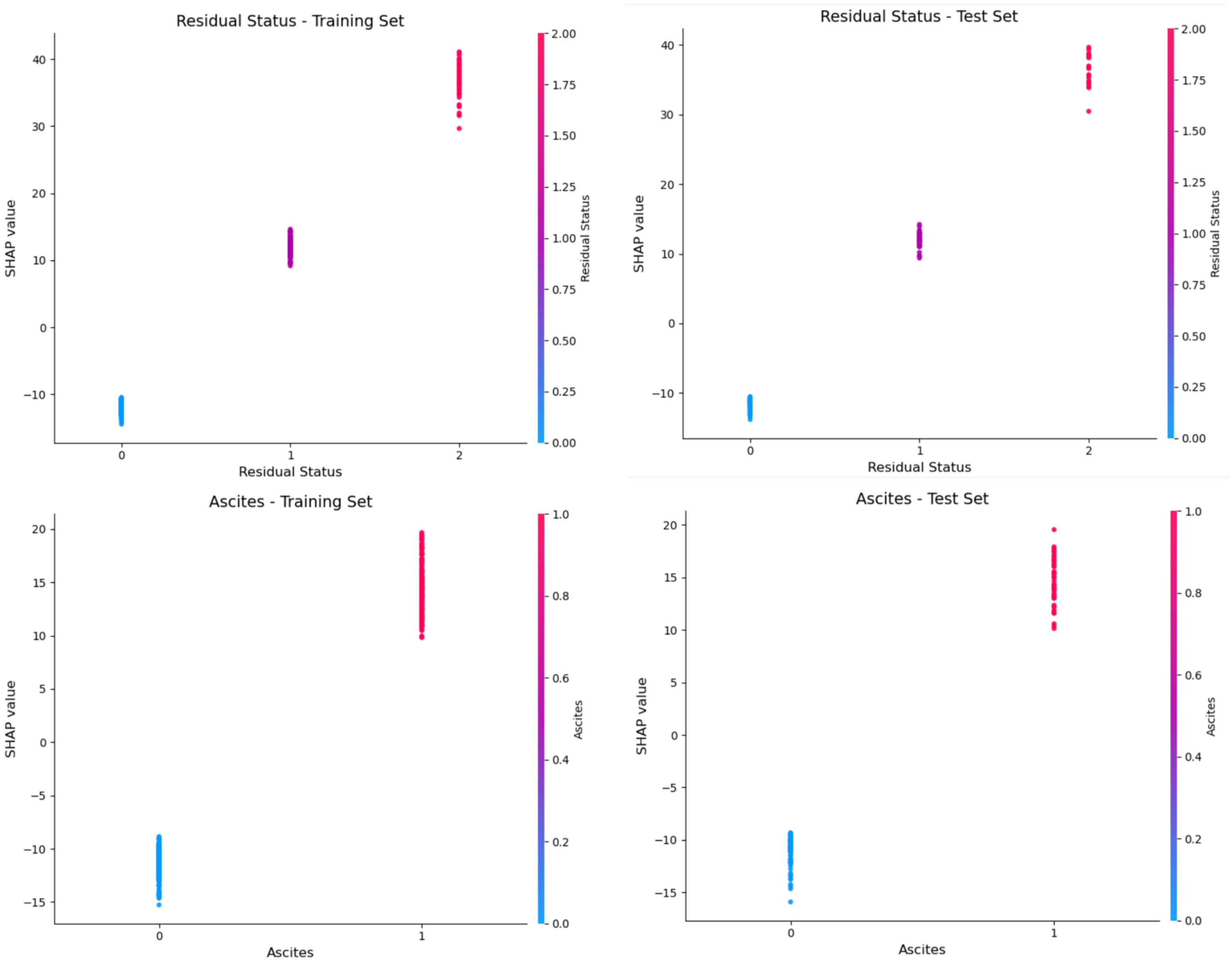


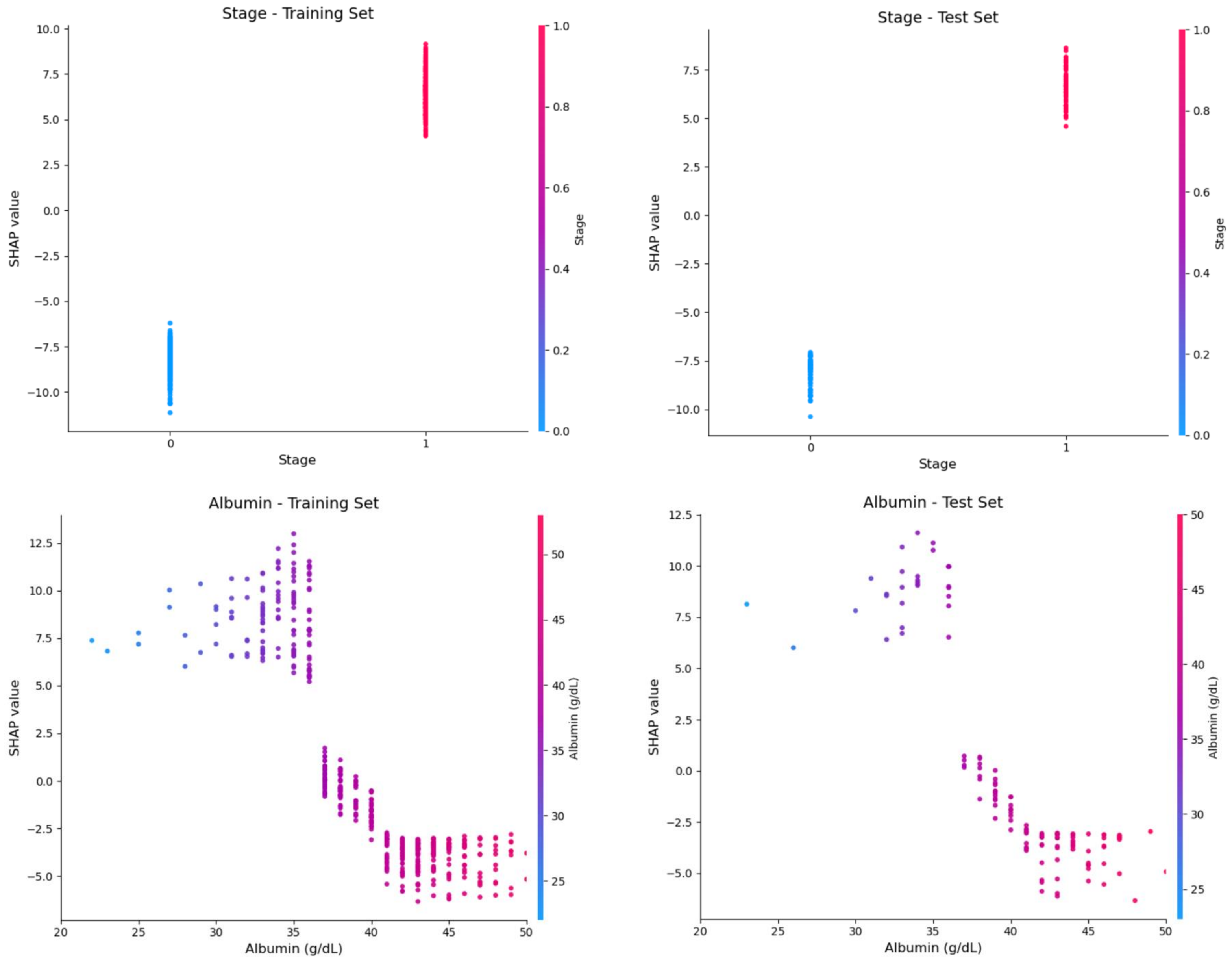
# SHAP Summary Plots



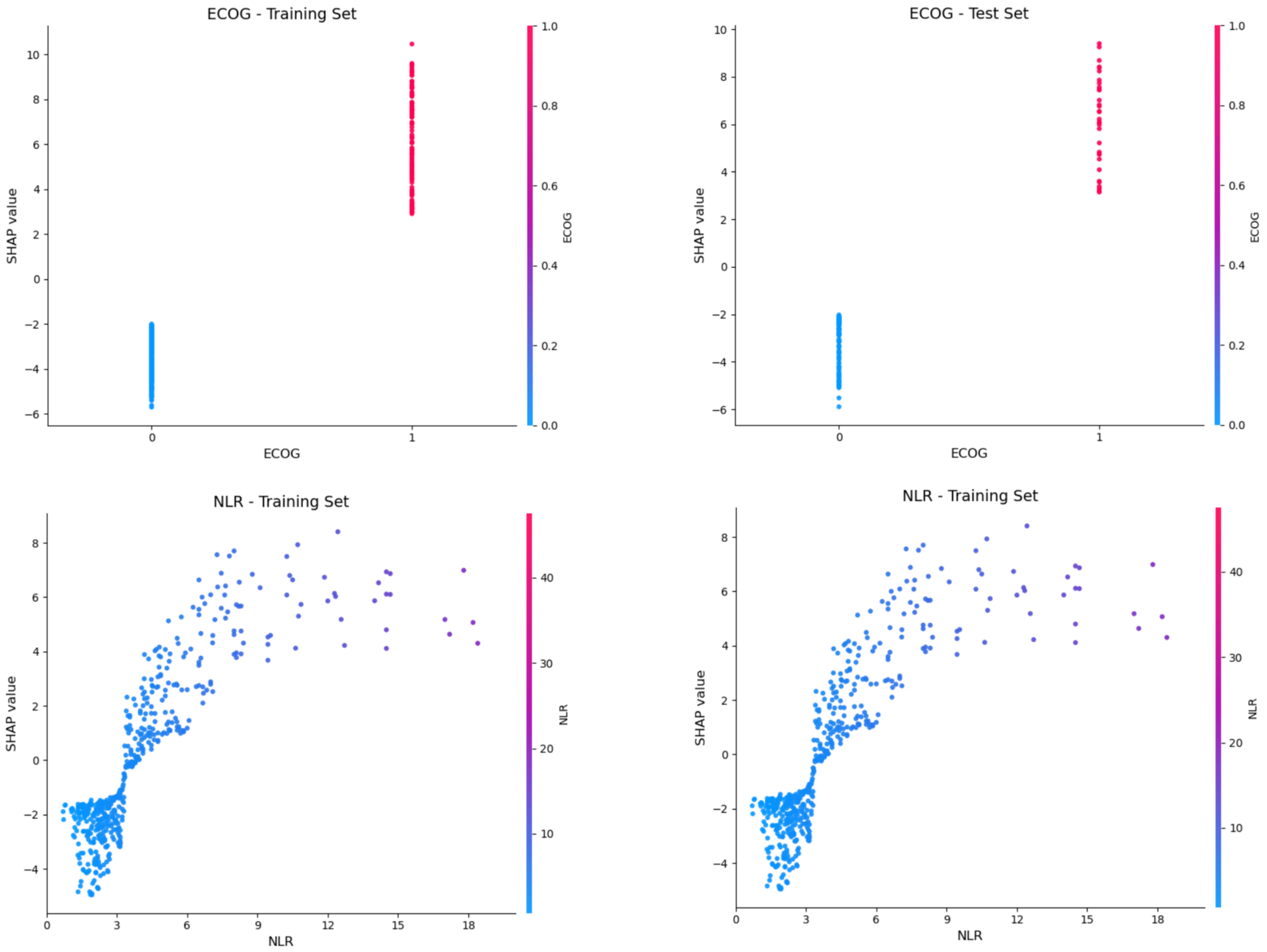



# SHAP Dependence Plots

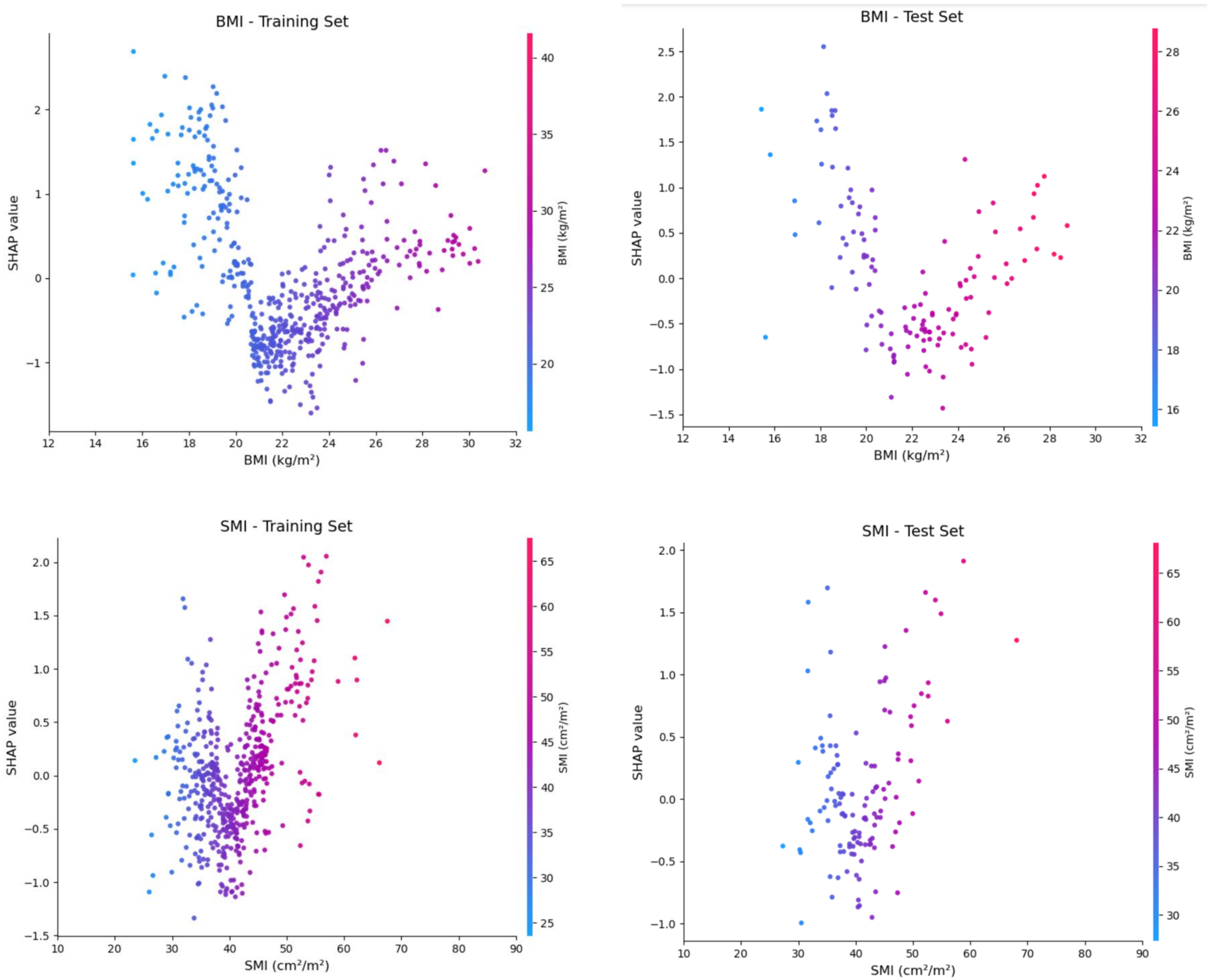




Stage - Training Set
SHAP value
Stage
Stage - Test Set
SHAP value
Stage
Albumin - Training Set
SHAP value
Albumin (g/dL)
Albumin - Test Set
SHAP value
Albumin (g/dL)



ECOG - Training Set
ECOG - Test Set
NLR - Training Set
NLR - Training Set
SHAP value
ECOG
NLR


BMI - Training Set
BMI - Test Set
SMI - Training Set
SMI - Test Set
SHAP value
BMI (kg/m²)
SMI (cm²/m²)

# 4. Conclusion & Discussion

## 4.1 Model Performance

- Cox and RSF showed the best and comparable discrimination.
- RSF achieved the highest mean time-dependent AUC and lowest 5-year Brier score.
- On the test set, paired bootstrap comparisons of C-index showed no statistically significant differences between models after Holm adjustment.

**Discussion**

- **Similar findings have been reported in lung (Germer et al., 2024) and colorectal cancer (Yang et al., 2023) survival studies, where machine learning and deep learning models generally failed to show significant prognostic superiority over traditional Cox regression.**
- Possible explanations include the **limited sample size and feature dimensionality, the absence of molecular or genomic predictors, and the use of clinically selected variables that had already been statistically associated with prognosis in previous studies**, leaving less additional prognostic information for more complex models to capture.

# 4. Conclusion & Discussion

## 4.2 Explainability

- **Cox:** Advanced stage, malignant ascites, and poorer ECOG status were the strongest predictors and were associated with increased mortality risk.
- **RSF:** Residual disease, malignant ascites, and advanced stage were the three most influential features, with higher values consistently increasing predicted risk.
- BMI showed a U-shaped risk pattern, whereas SMI had a relatively small effect and no clear risk cut-point was identified.

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# Thank you !